

ULJIN, Korea, Republic of

WMO: 471300

Lat: 36.9917N

Lon: 129.4128E

Elev: 51

StdP: 100.71

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-5.3	-23.3	0.5	-4.6	-21.1	0.6	-3.4	10.2	1.5	8.9	1.9	4.0	230	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	30.9	24.2	29.2	23.7	28.0	23.3	25.6	28.8	24.9	28.0	24.2	27.2	3.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.8	20.0	27.4	24.0	19.0	26.8	23.2	18.1	26.2	79.5	28.7	76.5	28.1	73.6	27.3	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.5	6.5	DB	-10.1	34.2	2.3	1.3	-11.7	35.1	-13.1	35.9	-14.3	36.6	-16.0	37.5	(4)
(5)			WB	-12.2	26.3	2.1	1.0	-13.6	27.1	-14.8	27.7	-16.0	28.3	-17.5	29.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	1.6	3.0	6.8	12.0	16.3	19.4	23.4	24.2	19.9	15.2	9.4	3.8	(6)	
	DBStd	8.32	3.26	3.51	3.72	3.88	3.31	2.65	3.08	2.59	2.37	2.82	3.58	3.62	(7)	
	HDD10.0	841	262	198	113	20	0	0	0	0	0	2	53	193	(8)	
	HDD18.3	2428	520	430	357	193	82	17	1	0	10	101	267	449	(9)	
	CDD10.0	1928	0	1	15	81	196	281	415	441	297	164	36	2	(10)	
	CDD18.3	473	0	0	0	4	19	48	157	182	57	5	0	0	(11)	
	CDH23.3	2754	0	0	2	26	96	163	1039	1238	185	7	0	0	(12)	
	CDH26.7	626	0	0	0	5	16	27	279	278	21	0	0	0	(13)	
(14) Wind	WSAvg	3.2	3.7	3.5	3.6	3.6	3.1	2.7	2.7	2.6	3.0	3.1	3.3	3.5	(14)	
(15) Precipitation	PrecAvg	1111	52	59	71	78	64	99	156	198	158	81	61	42	(15)	
	PrecMax	1551	141	180	168	185	150	203	326	513	348	230	191	200	(16)	
	PrecMin	684	0	6	4	24	11	11	37	23	9	6	3	0	(17)	
	PrecStd	243	43	45	40	44	40	51	82	125	93	62	53	46	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	14.7	20.9	26.3	28.9	29.7	33.3	33.7	29.1	24.1	20.5	15.3	(19)	
		MCWB	6.4	8.1	11.3	14.4	16.7	19.6	24.7	25.1	23.3	17.6	13.5	9.8	(20)	
	2%	DB	9.6	12.4	17.2	22.8	25.9	26.8	31.1	30.9	26.7	22.4	18.5	12.7	(21)	
		MCWB	4.5	7.1	9.6	13.2	15.6	19.7	24.2	25.1	22.2	17.2	13.0	7.7	(22)	
	5%	DB	8.1	10.5	15.0	20.2	23.8	25.1	29.5	29.3	25.3	21.2	16.9	11.1	(23)	
		MCWB	3.5	5.7	8.9	12.1	15.1	19.5	24.0	24.6	21.4	16.6	12.1	6.3	(24)	
	10%	DB	6.7	8.8	13.0	18.0	21.9	23.6	28.1	28.1	24.1	20.2	15.4	9.6	(25)	
		MCWB	2.6	4.5	7.9	11.3	14.8	19.2	23.6	24.1	20.6	16.0	10.9	5.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	10.8	13.2	16.9	19.0	22.6	26.3	27.0	24.6	20.6	16.1	11.2	(27)	
		MCDB	9.6	12.4	17.6	22.8	23.4	25.6	29.6	29.9	27.0	22.3	18.0	13.9	(28)	
	2%	WB	5.8	8.1	11.3	14.9	17.6	21.4	25.5	26.0	23.3	19.0	14.3	8.7	(29)	
		MCDB	8.2	11.1	15.4	19.6	22.2	24.3	29.0	29.1	25.9	20.9	17.0	11.8	(30)	
	5%	WB	4.4	6.4	9.8	13.7	16.7	20.7	24.7	25.3	22.2	17.8	12.9	7.1	(31)	
		MCDB	7.0	9.7	13.4	18.1	20.9	23.4	28.4	28.1	24.4	20.2	16.1	10.1	(32)	
	10%	WB	3.2	5.0	8.5	12.7	16.0	20.0	24.0	24.7	21.2	16.7	11.6	5.7	(33)	
		MCDB	5.9	8.1	12.4	16.3	19.8	22.4	27.6	27.4	23.4	19.4	14.5	8.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	7.9	8.1	8.2	7.4	5.5	4.9	5.1	6.5	8.2	8.5	8.3	(35)	
		MCDBR	8.8	9.7	11.0	11.2	10.8	8.0	7.5	7.0	7.7	8.9	9.6	9.4	(36)	
	5% WB	MCWBR	6.3	6.6	6.9	6.1	5.2	3.9	3.2	3.4	4.6	5.7	6.5	6.6	(37)	
		MCWBR	7.5	8.7	9.0	8.9	8.1	6.1	6.6	5.8	6.5	7.1	8.1	8.4	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.423	0.485	0.580	0.613	0.620	0.662	0.625	0.567	0.497	0.482	0.447	0.413	(40)	
		taud	1.994	1.874	1.696	1.681	1.708	1.685	1.830	1.969	2.097	2.006	2.019	2.035	(41)	
	Ebn at Noon	Ebn at Noon	719	720	688	698	703	673	695	726	757	716	688	698	(42)	
		Edh at Noon	134	171	224	239	237	242	208	177	148	147	130	121	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.48	3.17	4.13	4.89	5.39	5.05	4.42	4.35	3.80	3.36	2.49	2.24	(44)	
	RadStd	RadStd	0.20	0.35	0.39	0.35	0.52	0.44	0.61	0.51	0.44	0.33	0.33	0.18	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	N/A	N/A	-1.03	N/A	N/A	N/A	N/A	N/A	+92	+49

Nomenclature: See separate page